

CHAPTER 28

Multimedia Learning About Physical Systems

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Abstract

This chapter examines how people understand diagrams, animations, and multimedia presentations in order to learn about physical systems. Comprehension of these representations is analyzed within an information-processing framework that specifies the cognitive processes involved in understanding external displays, including attention, encoding, inference, and integration of different representations. The chapter first reviews the literature on comprehension of physical systems from diagrams (both static and animated) alone and then reviews how people understand physical systems from multimedia presentations in which diagrams are augmented by verbal instruction. The picture that emerges is that diagram comprehension is an active process of knowledge construction, rather than a passive process of internalizing the information presented in an external display. Because multimedia understanding depends on active information processing, it can be influenced considerably by the abilities, skills, and knowledge of the student. What is learned from a multimedia display is jointly determined

by aspects of the display and aspects of the learner.

Introduction

Diagrams accompanied by text have been a common means of representing and communicating information throughout history (Ferguson, 1992). In recent years, with advances in graphic technologies, innovations such as animations and interactive visualizations have made diagrammatic representations even more prevalent in scientific and technical discourse and in everyday life. These new media, and graphical displays in general, are believed to have enormous potential for education and training. But in order to realize this potential, we need basic research on how people comprehend and make inferences from diagrams and multimedia displays. The goal of this chapter is to review the literature on comprehension of static diagrams, animated diagrams, and multimedia (combining diagrams and verbal representations) from an information-processing perspective.

The chapter examines the representations and cognitive processes involved in understanding media, the abilities and skills on which this understanding depends, and the effectiveness of different media for communicating different types of content.

The chapter focuses on how people understand and learn about complex physical systems from different media. Physical systems include man-made mechanical systems and biological systems such as organs and organisms, that are made up of physical, visible subsystems that interact to accomplish some function, for example, stopping your car (in the case of car brakes) or pumping blood (in the case of the human heart). Most of the examples in this paper come from the domain of mechanics, specifically how mechanical devices work. However, I also briefly consider some other domains such as anatomy and geology.

Understanding a physical system involves several types of knowledge (Chi, de Leeuw, Chiu, & LaVancer, 1994; Hegarty & Just, 1993; Narayanan & Hegarty, 1998). First, it involves knowledge of the parts of the system, what they are made of, and how they are connected and configured in space, that is, the *configuration* of the machine. This knowledge corresponds to a static mental model of the machine. Second, it includes knowledge of how the components move and affect each other's motions, that is, the *behavior* of the machine. Central to this is knowledge of the causal chain or chains of events in the machine's behavior. It includes both kinematic understanding of how the different machine components move in space and affect each other's movements, and dynamic understanding of the forces that bring about these movements. This knowledge corresponds to a dynamic mental model of the machine. Finally it involves understanding the *function* of the machine, that is, what it is designed to do, and how the structure and behavior achieve this function.

In the first section of this chapter I present a theoretical framework that specifies the cognitive processes involved in understanding and learning about physical systems from

multimedia presentations. In the second section, I discuss the different types of external representations (diagrams, animations, texts, etc.) that are used to represent physical systems in multimedia presentations. The third section addresses how and to what extent people can understand systems from diagrams (both static and animated) alone. Finally in the fourth section, I consider how people understand physical systems from multimedia presentations, that is, how they understand verbal information, in conjunction with static and animated diagrams, in order to construct mental models.

A Theoretical Framework for Understanding Multimedia Comprehension

Figure 28.1 represents the situation when someone has to understand a mechanical system from a single medium or multimedia display. The display is made up of text, diagrams, and so forth that are *external representations* of some system. The goal of comprehension is to construct an *internal representation* of that system, which can also be called a *mental model*. The information in the external display is perceived by the senses – vision (for text, diagrams, and animations) or audition (for commentaries). The user is not passive in this process, but actively attends to information in different locations in the display or in different modalities to encode the information at that location or in that modality. These basic attentional, perceptual, and encoding processes result in an internal representation of at least some of the information presented in the multimedia display.

In addition, to these basic perceptual, attentional, and encoding processes the user must integrate the information presented at different locations in the display and in different media and modalities to construct an internal representation or mental model of the referent of the text. This may include translating information from a verbal to a spatial code, or vice versa (e.g., to integrate

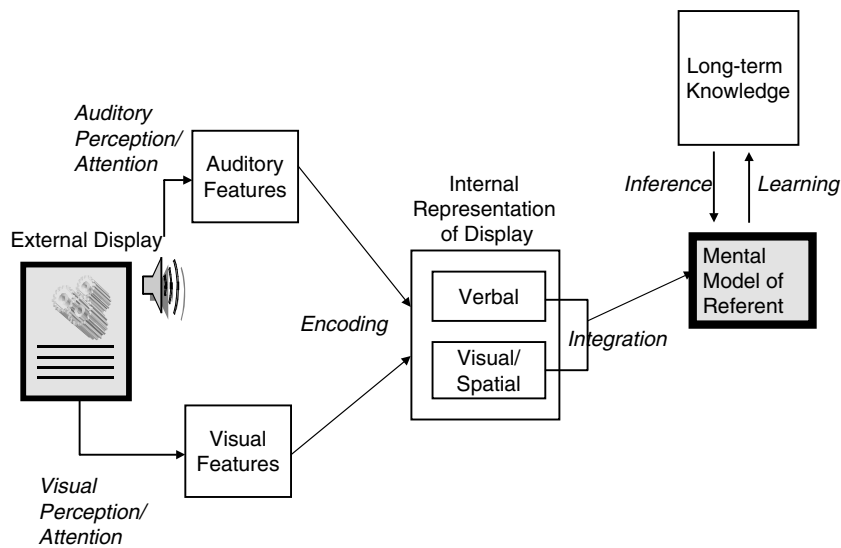


Figure 28.1. An information-processing model showing the different representations (external and internal) and cognitive processes involved in the comprehension of multimedia.

information in a text and diagram). In constructing this mental model, the user might also make some inferences from the information presented in the display, based on his or her prior knowledge or other inference mechanisms (e.g., visualizing how some object moves). These inference processes involve going “beyond the information given,” that is constructing new knowledge, rather than merely making an internal copy of the information that is presented externally in the multimedia display. If inference processes occur, the resulting internal representation will contain information that is not presented explicitly in the external display.

The comprehension of a display therefore depends not just on qualities of the display itself (e.g., what information it contains and how this information is represented externally), but also on the quality of the comprehension processes that operate on the external representation, including perception, attention, encoding, integration, and inference. For example, if the user does not attend to all of the relevant information, his or her internal representation may contain less information than the external display. If he or she has limited working memory capacity, this might affect his or her ability to integrate

information that is presented in different locations or at different times. If he or she has a great deal of background knowledge about the topic of the display, he or she may draw inferences from the information presented that would not be part of the comprehension of a less knowledgeable user. Thus, the internal representation that results from comprehension might be quite different in content and format than the external representation that gave rise to this representation.

Representations of Physical Systems in Different Media

We now consider the main types of external representations used to communicate about physical systems, and the characteristics of these types of representations. A representation is something that stands for something else and can exist either in the world as an external representation, or in the mind of a person as an internal representation. The representations included in multimedia displays include various types of diagrams and verbal materials. To understand the nature of these representations, we must specify the nature of the object represented (i.e., the

referent), the representational medium, the correspondence between these two, and the cognitive processes that operate on the representation (Markman, 1999; Palmer, 1978). Because diagrams and verbal materials are fundamentally different types of representations, they depend on different comprehension processes, and may be optimally suited to conveying different types of information.

Static Diagrams

Diagrams are spatial representations, that is, visual-spatial arrays, in which information is communicated by spatial properties such as shape, location, and adjacency of parts. Physical systems are also essentially spatial entities because their operation depends on the shape, connectivity, and movement of their components. Although spatial representations can represent both spatial and nonspatial information (e.g., graphs represent number), in the mechanical domain they typically represent spatial properties (Hegarty, 2004). Thus, most diagrams of physical systems are *iconic*, in the sense that spatial relations between objects in the representation correspond to, or are *isomorphic* to the spatial relations between the objects that they represent. Figures 28.2, 28.3, and 28.4 present examples of some types of diagrams that are typically used to communicate about machines. Note that an iconic diagram does not necessarily represent all of the visual features of its referent. For example, in Figure 28.1, pulleys are represented by circles, thus omitting a great deal of the visual detail of what a pulley looks like. This demonstrates another important characteristic of representations, which is that they can be schematic, in the sense that they represent some aspects of the referent but not others. Typically in instructional materials about machines, the important spatial relations between the parts of the machine are preserved in the diagram.

Static diagrams by definition do not move. However movement is an essential property of mechanical systems. Therefore understanding how a machine works from a static diagram involves the ability to infer the motion of the device from the information

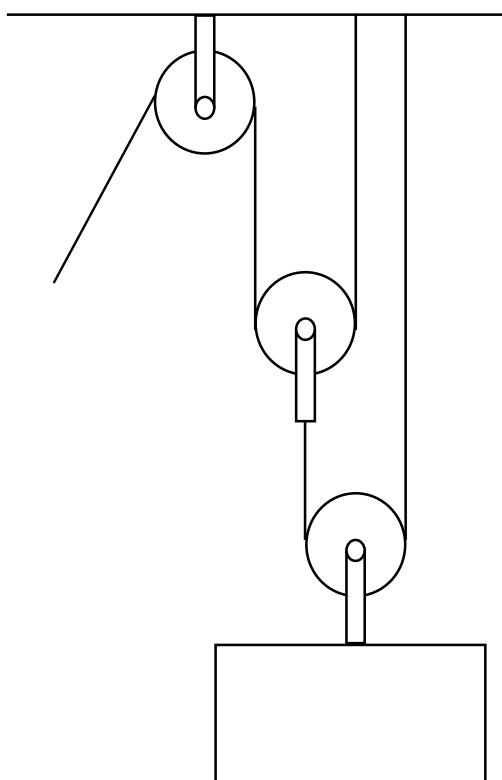


Figure 28.2. Schematic diagram of a pulley system.

given in the static diagram. In this chapter, I refer to the process of inferring motion from static displays as *mental animation* (Hegarty, 1992).

Engineers and graphic designers have developed several different graphic conventions to convey information about movement in static diagrams (Ferguson, 1992). Two common conventions are to annotate

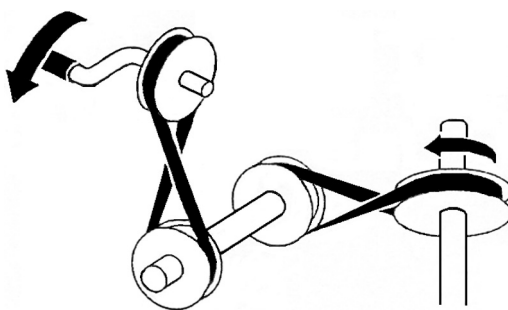


Figure 28.3. Diagram of a belt and pulley system, annotated by arrows. An example of a spatial mechanism in which the mechanical interactions are in different planes.

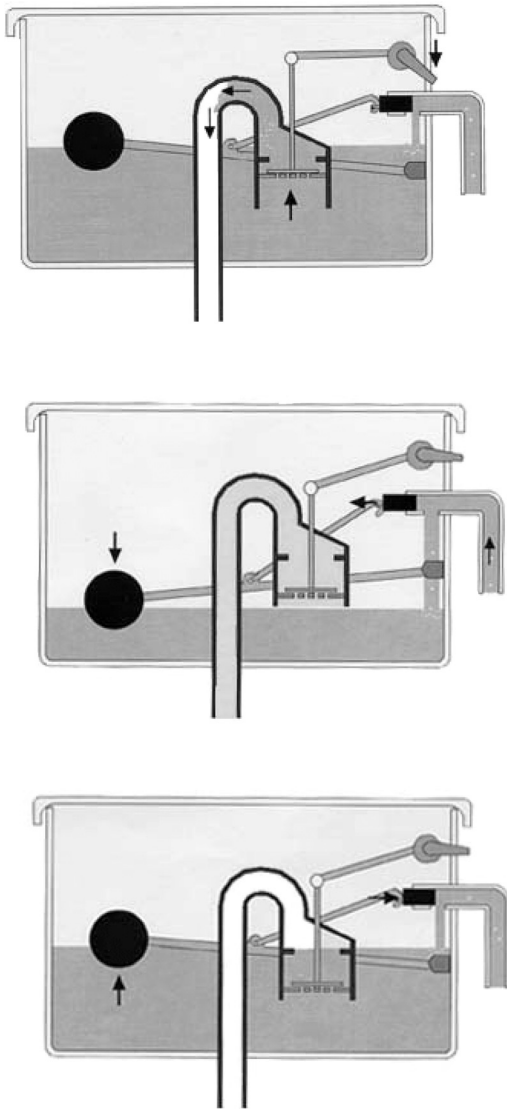


Figure 28.4. Three phase diagrams depicting different stages in the operation of a flushing cistern mechanism (this mechanism is often used to flush toilets) used by Hegarty et al. (2003) (Figure reprinted by permission of Lawrence Erlbaum Associates).

diagrams with arrows, (as in Figure 28.3) and to present a series of phase diagrams or “small multiples” (Tufte 1997), showing the configurations of the components of a mechanical device at different phases of its operation (as in Figure 28.4). When static diagrams are used to convey motion in this way, the information about how the machine moves is not directly represented in

the diagram. In these situations, developing a mental model of the movement of the machine still involves some type of inference, such as inferring what movement causes the state shown in one phase diagram to be transformed into the state shown in another phase diagram, or interpreting arrows to imagine how the parts of a machine move to achieve its function.

Animated Diagrams

The animated diagram is a relatively recent graphic invention. An animation is essentially a sequence of static images, that when played at a relatively rapid rate, produces the effect of apparent motion. Animations allow us to represent the motion of a mechanical system directly (rather than through the more indirect means of arrows and phase diagrams). In this sense we can think of them as more *realistic* representations, that is, more isomorphic to the reality that they represent. Comprehension of how a machine moves from an animation depends on the ability to perceive the motion of the system (rather than to infer this motion).

Like static diagrams, animated diagrams can be more or less isomorphic to reality. They can show processes in real time, in slow motion, or speeded up (e.g., animations can take a few seconds to show geological phenomena that occur over millennia). Animations can also slow down critical phases of a process (that occur very quickly in reality) while speeding up others. In these cases, animations depart somewhat from reality. Animations can vary in their interactivity, that is, the extent to which the user has control over the speed and sequence of frames. A common method of adding interactivity to an animated diagram is to use a “slider bar,” which allows users to move forward and backward through the animation at their own pace and in any order.

Finally, in addition to depicting motion, another common use of animated diagrams is to show the structure of a mechanical or biological system that cannot be easily depicted in a single two-dimensional diagram. Mechanical systems can be classified as *planar* or *spatial* mechanisms (Dar,

Joskowicz, & Rivlin, 1999). Planar mechanisms are those in which the mechanical interactions take place in a single plane (e.g., the pulley system shown in Figure 28.2 and the flushing cistern shown in Figure 28.4). Spatial mechanisms are those in which the mechanical interactions occur in three dimensions (e.g., the belt and pulley system shown in Figure 28.3). Biological structures, such as ball-and-socket joints, the heart, and lungs are spatial mechanisms. In spatial mechanisms it is not always possible to show all of the important mechanical interactions in a two-dimensional diagram. In these cases, an animation that shows the mechanism rotating in place can be used to show the three-dimensional structure of the device. Understanding such an animation involves the ability to integrate the sequence of two-dimensional views from different spatial locations to construct an internal representation of the three-dimensional structure of the device.

Verbal Materials

Verbal materials are descriptive representations, which express facts or assertions about their referents, depend on arbitrary symbols (i.e., words), and have a linear structure. Unlike diagrams, verbal representations are not isomorphic to the reality that they represent. For example, consider the following text describing the pulley system shown in Figure 28.2 (Hegarty & Just, 1993).

The system consists of three pulleys, two ropes and one weight. The upper pulley is attached to the ceiling. The other pulleys are free to move up and down. The upper rope is attached to the ceiling at one end, goes under the middle pulley and over the upper pulley and is free at the other end. . . .
(p. 721)

Understanding this text involves understanding the meaning of words such as *pulley* and *rope*, as well as using knowledge of the syntax and semantics of English to construct an internal representation of the configurational relations between the parts of the device. In contrast to diagrams, verbal information has a linear structure, such that the order of

words in a sentence and sentences in a discourse communicate meaning. In this sense it might be particularly suited to describing information that has a linear structure, such as a sequence of events.

Verbal information can be presented either as visual text or as an aural commentary. These involve some inherent differences. For example, reading text is self-paced in that text is static on the printed page, and the reader can slow down, speed up, or reread as often as he or she wishes in order to understand the text. In contrast, an aural commentary is heard at a particular rate, corresponding to the normal rate of speech. It is possible to make an aural commentary somewhat interactive, such that the user can replay sections, and so forth. However one cannot arbitrarily slow down or speed up the rate of speech and maintain comprehension.

Relations Between Diagrams and Verbal Instruction

Multimedia displays present information in both verbal and visual-spatial codes. The verbal and visual-spatial information can occur in different relations to each other. In terms of content, verbal and spatial information can be redundant (e.g., when a text describes the structure of an object that is also depicted in a diagram). Verbal and spatial information can also be complementary (e.g., in the case where a diagram shows the static structure of a mechanical system and a text describes how parts of the machine are free to move). In addition to these semantic relations, verbal and spatial information can have different relations as they are presented. For example, with printed media, the text can be either separate from the diagram, or integrated with the diagram and with animated media. A verbal commentary can occur either simultaneously with the animation that it explains or before or after this animation.

Finally, in addition to diagrams and verbal information, there are several other sources of information that may occur in multimedia displays, such as nonverbal sound effects (Moreno and Mayer, 2000) and haptic

feedback. These are beyond the scope of this chapter, which will focus on static and animated diagrams and verbal materials presented in the visual and auditory modalities.

Understanding and Learning from Diagrams

In this section we will review how people learn about the structure and functioning of physical systems from diagrams alone, including static and animated diagrams. One might wonder why we should focus on comprehension of diagrams alone in a handbook of multimedia. That is, why should we examine comprehension of diagrams alone, without accompanying verbal instruction, when it has often been shown that people learn more from diagrams that are accompanied by explanatory text or verbal commentaries (e.g., Hegarty & Just, 1993; Mayer & Anderson, 1991; Mayer & Gallini, 1990). However, if our goal is a scientific account of how multimedia presentations are understood, it is important to specify the cognitive processes and comprehension outcomes that result from exposure to the individual media that make up multimedia presentations. This scientific account, in turn, can be the basis of a set of principles for the design of multimedia presentations (cf. Narayanan & Hegarty, 1998; 2002), for example by specifying when and how diagrammatic materials should be augmented by text.

Understanding Static Structure from Static Diagrams

An internal representation of the configuration of a physical system, (i.e., its static structure) is an important component of a mental model. Multimedia presentations typically include iconic diagrams to show the structure or configuration of parts of a system, including the shapes of components, how they are configured in space and how they are connected. In terms of the information-processing analysis in Figure 28.1, understanding the structure of a mechanical system from a static iconic diagram is essen-

tially a process of encoding the information in the external representation. Some studies have shown that people are quite good at this process. For example, Hegarty and Just (1993) found that showing people a static diagram (without accompanying text) was sufficient for understanding the structure of a pulley system. Similarly, when Heiser and Tversky (2002) presented students with static diagrams of different mechanical systems and asked them to describe the diagrams, their participants spontaneously described the structure of the systems.

However, even iconic static diagrams can be highly conventionalized, so that constructing a mental model of the structure of a system from a static diagram is not necessarily a trivial comprehension task. Two types of diagrams that have been shown to present problems for students are schematic diagrams and two-dimensional diagrams that depict three-dimensional objects.

SCHEMATIC DIAGRAMS

Highly schematic diagrams, which omit visual details of the appearance of their referents, can sometimes be misunderstood. Ferguson and Hegarty (1995) examined how students reasoned about pulley systems based on static diagrams or three-dimensional models (made up of actual pulleys and ropes). Like the example in Figure 28.2, the diagrams were highly schematized, for example, they used circles to represent pulleys and lines to represent ropes. Ferguson and Hegarty found no effect of instruction medium (schematic diagrams vs. real models) on how well students could learn to compare the efficiency of different pulley systems. However, when asked to apply their knowledge to novel tasks, students who learned from real models had superior performance. Furthermore, students' drawings indicated that those who learned from schematic diagrams did not understand the structure of the systems (i.e., the spatial relations between the ropes and pulleys). Schwartz (1995) also found differences in the types of mechanical reasoning afforded by realistic and schematic diagrams, such

that more realistic diagrams induced students to simulate how the mechanical systems moved and more schematic diagrams were associated with more analytic strategies. These studies indicate that understanding the structure of a physical system from a diagram is not always a trivial comprehension task, and that more and less schematic diagrams may naturally evoke different inference processes.

DIAGRAMS OF THREE-DIMENSIONAL OBJECTS

Diagrams of three-dimensional objects can also cause comprehension difficulties. Because the printed page and computer screen are essentially two-dimensional entities, it is impossible to directly represent a three-dimensional object in these media. Any diagram of three-dimensional objects must involve some projection of three dimensions onto the two-dimensional plane. Graphic designers over the centuries have invented a number of graphic conventions to display three-dimensional structure in two-dimensional media, such as perspective views, multiple orthographic views, and cross-sections. Constructing a static mental model from these types of diagrams involves such processes as integrating different two-dimensional representations into a single three-dimensional internal representation, or visualizing the internal structure of an object. Students, particularly those with low spatial abilities, have difficulty both producing and comprehending such diagrams (Cohen, Hegarty, Keehner, & Montello, 2003; Eley, 1983; Kali and Orion, 1996).

Understanding Structure from Animated Diagrams

A recent solution to displaying three-dimensional structure in two-dimensional displays is to show the object rotating in depth on a computer screen. In this type of display, time is used to show the third dimension of space and the display exploits movement-based depth cues such as motion parallax to give a strong impression of a three-dimensional object in a two-dimensional display. Many educators believe

that these types of displays may be particularly effective in overcoming students' limitations in understanding three-dimensional structures from two-dimensional displays. However, empirical research on use of animations to teach anatomical structures suggests that comprehension of these displays may be even more dependent on spatial abilities. In a study where multiple views of three-dimensional anatomy were presented to medical students using a rotating computer model, a subsequent test of anatomical knowledge showed a significant disadvantage to individuals with poor spatial abilities (Garg, Norman, Spero, & Maheshwari, 1999). For these students, learning was effective only if the display was restricted to a simple depiction entailing just two cardinal views, suggesting that complex three-dimensional computer visualizations may actually impair spatial understanding for low spatial individuals. The effects of spatial ability, however, can be moderated by the characteristics of the computer simulation. The performance of low spatial medical students has been enhanced, for example, by allowing them to direct the rotation of the model, suggesting that learner control contributes significantly to the successful integration of complex spatial information (Garg, Norman, & Sperotable, 2001). Other variables that might potentially enhance or diminish understanding include the complexity of the image, the depth cues available, and the type of interface used to interact with the animation. In order to guide the future evolution of three-dimensional visualizations, these factors need to be explored and the optimum combination of features established.

Understanding Dynamic Events from Static Diagrams

Understanding physical systems such as machines includes an understanding of how the parts move and constrain each other's movements to achieve the function of the system. Achieving this type of understanding from a static diagram depends on "mental animation," that is, the inference of kinematics

(how objects move) from their static structure. There is little evidence that people do this spontaneously when they view a static diagram. For example, people have poor comprehension of the kinematics of a device after they study a static diagram (Hegarty and Just, 1993) and when asked to describe a static diagram of a mechanical system, they do not spontaneously describe how the parts move (Heiser & Tversky, 2002). In other words, viewing a static diagram of a mechanical system alone does not induce people to think about the kinematics of the device.

But can people mentally animate static diagrams when a cognitive task demands it? There is considerable evidence that they can, to some extent. Hegarty (1992) presented students with static diagrams of pulley systems, such as the one presented in Figure 28.2, and asked them to verify sentences such as “when the rope is pulled, the middle pulley turns clockwise.” Many participants were readily able to make this inference. Similarly Schwartz and Black (1996a) found that people could accurately infer the direction of rotation of gears in a gear chain and further research on mental animation has generalized these results to the inference of motion in a variety of mechanical systems (Hegarty & Kozhevnikov, 1999; Hegarty & Steinhoff, 1997).

Research on mental animation has provided important insights into the nature of people’s internal representations of mechanical systems. Although mechanical systems in motion involve continuous motion over time, people think of these systems in terms of a causal sequence of events. Hegarty (1992) measured errors, reaction times, and eye fixations as people inferred the motion of different components in a mechanical system. I found that reaction time increased and accuracy decreased when people inferred the motion of components later in the causal chain of events compared to components earlier in the chain of events. When asked to verify the motion of a component, say the middle pulley in Figure 28.2, people looked at that component, and components earlier in the causal chain of events (i.e., the up-

per rope and pulley), but not components later in that chain of events. These data indicate that people infer the motion of components piecemeal, beginning by imagining the rope being pulled, and working through the causal chain of events in the motion of the system.

Research on mental animation has also provided evidence for spatial imagery as a strategy in mental animation (Hegarty, 2004). For example, Schwartz and Black (1996b) found that when people inferred the rate of rotation of two interlocking gears (whether a knob on one gear would mesh with a groove on a second gear), response time was proportional to the angle of rotation, as in studies of mental rotation, suggesting that people were using an analog imagery strategy to infer how the gears would rotate. However, this does not mean that mental animation is a process of inspecting a holistic detailed visual image in the “mind’s eye.” As we have seen, mental animation involves imagining the motion of mechanical systems one component at a time (in order of the causal chain of events), and although this is sometimes an analog imagery process, the inference can also be based on verbally coded knowledge, for example, “every other gear in a gear chain turns in the same direction” (Schwartz & Black, 1996a).

Finally, although a useful strategy in understanding dynamic systems from static images, mental animation ability is limited. The research on naïve physics (Kaiser, Proffitt, Whelan, & Hecht, 1992; McCloskey, 1983) has revealed several systematic errors in people’s inference of motion from static diagrams, for example the classic naïve intuition that a ball shot through a curved tube will emerge in a curved path (McCloskey, 1983). Furthermore low spatial-ability individuals are less accurate in inferring how mechanical systems move from static diagrams (Hegarty & Kozhevnikov, 1999; Hegarty & Sims, 1994; Hegarty & Steinhoff, 1997). The research on mental animation has therefore revealed that people have some ability to infer how a machine moves from a single static diagram, but that this ability is limited, especially for people with low spatial ability.

GRAPHIC DEVICES FOR COMMUNICATING DYNAMIC EVENTS

As shown in Figures 28.3 and 28.4 graphic designers often communicate motion in static diagrams by annotating parts of the diagrams with arrows, or by presenting a series of phase diagrams showing the configuration of parts of a system at different phases of its motion. There have been relatively few studies that assessed the effectiveness of these graphic devices. Heiser and Tversky (2002) found that when people were asked to describe static mechanical diagrams that included arrows showing the motion of system parts, they spontaneously provided functional descriptions of how the system worked. In contrast, if the diagrams did not include arrows, people described only the structure of the system. Hegarty, Kriz, and Cate (2003) examined people's comprehension of a flushing cistern after they viewed the three static diagrams shown in Figure 28.4. People who viewed these phase diagrams had better understanding of the mechanical system compared to those who viewed a single static diagram. These studies indicate that arrows and phase diagrams are promising ways of communicating the kinematics of a device. Because these diagrams are static, understanding kinematics from these diagrams still depends on mental animation. However, arrows and phase diagrams may be useful ways of aiding this mental animation process, especially for low spatial-ability students and when explaining relatively complex mechanical systems.

Understanding Dynamic Events from Animated Diagrams

On an intuitive level, animations seem like an obvious way to communicate dynamic events, such as how a machine works. Animations portray the temporal changes in the operation of a machine explicitly so that the external display (i.e., the animation) can be isomorphic in both space and time to the dynamic process that it is intended to represent. In terms of the information-processing analysis presented in Figure 28.1, learning how a complex physical system works from

an animation relies on the ability to *perceive* the motions of components, whereas learning from static diagrams depends on the ability to *infer* these motions.

It is therefore initially surprising to most people that there has been very little (if any) evidence that animations are more effective than static diagrams for communicating about dynamic processes. For example, in the study by Hegarty et al. (2003) students learned how a mechanical system (i.e., a flushing cistern) worked from either a single static diagram, the three phases diagrams shown in Figure 28.4, or an animation. We found that although the three phases diagram group and animation group had a better understanding than those who viewed a single diagram, there was no significant difference in comprehension between the three phases diagram group and the animation group. It appears, therefore, that participants were able to interpolate between the three phases shown in the static diagrams (i.e., 3 of the 140 frames included in the animation) to mentally animate their internal representations of the device, and this enabled them to understand the device as well as people who actually viewed all 140 frames of the animation.

It might be tempting to conclude that the animation used by Hegarty et al. (2003) was not a good one, or that the flushing cistern is a particularly difficult device to learn about from an animation. However, Narayanan and Hegarty (2002) found similar results when we used stimuli from the award-winning *The Way Things Work* book and CD-ROM (Macaulay, 1988, 1998). Furthermore, in a review of over 20 studies that compared learning from static and animated graphics in a variety of domains (e.g., Hegarty et al., 2002; Hegarty, Quilici, Narayanan, Holmquist, & Moreno, 1999; Narayanan & Hegarty, 2002; Palmiter & Elkerton, 1993; Palmiter, Elkerton, & Baggett, 1991; Pane, Corbett, & John, 1996; Rieber, 1989; Rieber & Hannafin, 1988) Tversky, Morrison, and Betrancourt (2002) concluded that there was no advantage of animations over static graphics. Although a small number of studies (e.g., Park & Gittelman, 1992;

Rieber, 1990, 1991) seemed to show such an advantage, on closer inspection, Tversky et al. (2002) found that in these studies the animations presented more information than the static diagrams, so that the two media were not informationally equivalent (cf. Larkin & Simon, 1987).

ARE ANIMATIONS MORE EFFECTIVE FOR LOW-SPATIAL STUDENTS?

One plausible hypothesis is that viewing animations might be more effective than static diagrams in the case of individuals with low spatial abilities, because these individuals have less ability to mentally animate static diagrams. If this is true, an animation might function as a type of cognitive “prosthetic” for those with low spatial ability. However, in a review of research that examined the effects of spatial ability on multimedia, Hegarty (2004) found no evidence for this hypothesis. In contrast, the literature indicated that in some cases (Hegarty et al., 2003; Narayanan & Hegarty, 2002) there was no aptitude treatment interaction (ATI) between spatial ability and format of diagrams (static versus animated) whereas in other cases, high spatial-ability individuals were better able to perceive and learn from animations (Garg et al., 1999; Isaak & Just, 1995; Mayer & Sims, 1994).

WHY ARE ANIMATIONS NOT MORE EFFECTIVE?

How can we explain why animations are not more effective than static diagrams in instructing people about dynamic processes. There are several plausible reasons. First, an important characteristic of animations is their transience (Ainsworth & Van Labeke, 2004; Stenning, 1998). In all of the experiments discussed in the preceding text the animations were computer controlled and played at a constant rate. When one views such an animation, one frame is seen at a time, and once the animation or video has advanced beyond a given frame, that frame is no longer available to the viewer. There is a question, therefore, of whether perception and comprehension of an animation can keep up with the pace at which the anima-

tion is presented, especially when different components of motion of the same object (e.g., rotation and translation), or the motions of different objects must be mentally integrated to understand the process being conveyed (cf. Isaak & Just, 1995). In contrast, viewing of static diagrams is self-paced and interactive in the sense that one can view a diagram for as long as one wishes or look back and forth between different phase diagrams as often as necessary to understand the dynamic process.

Second, visual attention might limit what is learned from an animation. In complex machines, several components move at once. For example, in the case of the flushing cistern, there are two chains of events that overlap in time; one that flushes water out of the cistern and the other that refills the cistern. To understand an animation of a complex machine, people need to attend to and relate changes that occur simultaneously in different regions of space. However, as recent research on change blindness has made clear, when our visual attention is focused on one location in a display, we are likely to be “blind” to changes in other regions of the display (Rensink, 2002). This will be particularly true when the most salient motions in the animation are not those that are most thematically relevant (Lowe, 1999, 2003).

Third, learning from animations might be less effective because viewing an animation is a passive process. In contrast, inferring how a system moves from a static diagram or series of phase diagrams (i.e., mental animation) is a more active cognitive process. Phenomena such as the self-generation effect (Slamecka & Graf, 1978) and the self-explanation effect (Chi et al., 1994) suggest that people learn more effectively if they are more active in the learning process, and generate ideas or explanations.

Fourth, there may be a mismatch between the format of an animation and the format of internal mental models of how machines work. When a machine is in motion, many of its components move at once, so a realistic animation of a machine shows these simultaneous motions. In contrast, when people attempt to understand

how a machine works, or answer questions about how it works, they infer the motion of components one by one, in order of a causal chain of events (Baggett & Graesser, 1995; Hegarty, 1992; Hegarty & Sims, 1994; Narayanan, Suwa, & Motoda, 1994). That is, they represent a sequence of events (cf. Zacks & Tversky, 2001) rather than continuous motion.

CAN WE MAKE ANIMATIONS MORE EFFECTIVE?

Can understanding the limitations of animation comprehension help us design more effective animations? Tversky et al., proposed two principles for the design of effective instructional materials, the *apprehension principle* and the *resemblance principle*. The apprehension principle is that external representations should be designed such that they can be “readily and accurately perceived and comprehended” (p. 258). For example, a violation of this principle is to present animations so fast that that perception and comprehension processes cannot keep up with the speed of the animation. The resemblance principle is that animations should be designed to correspond to the type of internal representation that one wishes to foster.

With respect to the apprehension principle, many researchers are currently exploring the use of interactive animations to communicate how physical systems work (e.g., Lowe, 1999; 2003; Schwan & Riempp, 2004). In interactive animations, viewers can speed up or slow down the display to match their comprehension speed, and can view and review parts of the animation in any sequence, skipping over parts that they understand while slowing down when they get to more difficult sections. Thus, the learner can tailor the instruction to his or her own needs. Furthermore, when interacting with dynamic displays, viewers are much more active in learning.

Interactivity raises its own difficulties for comprehension, however. First, interactive animations must have an interface, and using an interface can take the viewer’s attention away from the task of understanding and learning from the animation. Second, to be effective, interactive displays assume

a learner who is not just motivated, but who has the metacognitive skills to use the interactivity effectively and the domain knowledge necessary to identify the most relevant aspects of the display. This is not always true in the case of novices. For example, Lowe examined how students interacted with an animation that displayed changes in weather conditions across the Australian continent over a seven-day period. In one study (Lowe, 1999) they were required to generate a verbal description of the animation, whereas in another (Lowe, 2004) they were asked to apply what they learned to predicting how weather conditions in another map would change over a 24-hour period. In both studies, students tended to pay attention to perceptually salient changes in the display and often ignored changes that were highly relevant to the meteorological tasks, but less conspicuous in the displays. Studies such as this indicate that interactive displays may not always be more effective than static displays especially for low knowledge students. Consistent with this view, Kriz and Hegarty (in press) found that students with low prior knowledge (humanities graduate students) were unable to construct accurate models of how a novel device works using interactive animations, whereas students with high domain knowledge (engineering graduate students) were able to construct the correct model.

There have been fewer studies addressing Tversky et al.’s (2002) apprehension principle, suggesting that animations should be designed to correspond to the type of internal representation that one wishes to foster. In this regard, studies of mental animation suggest that the internal representation of a dynamic physical system is of a causal chain of events, rather than a continuous animation (Hegarty, 1992). Therefore in accordance with Tversky et al.’s *resemblance principle*, it might be more effective to communicate how a machine works using a sequence of static phase diagrams that show the beginning and end points of the important causal events, and preliminary findings (Hegarty et al., 2003) are consistent with this prediction.

Understanding and Learning from Multimedia Presentations

In the final section of this chapter, I consider learning from truly multimedia presentations, that is, those that include both diagrammatic representations (static or animated) and verbal instruction. We have seen that although diagrams can be very effective in communicating about dynamic causal systems, they also have limitations, especially for learners with poor spatial ability and low domain knowledge. For example, the studies reviewed in the preceding text indicate that these individuals have difficulty understanding highly schematized diagrams, understanding the three-dimensional structure of an object from two-dimensional representations and mentally animating static diagrams. It is not surprising, therefore, that several studies have shown that people learn more from diagrammatic representations if they are accompanied by verbal instruction (e.g., Hegarty & Just, 1993; Mayer & Anderson, 1991; Mayer & Gallini, 1990). To better understand these results, I now consider the possible relations between verbal and diagrammatic representations in multimedia presentations, including content relations and format relations, and how these affect comprehension.

Content Relations Between Diagrams and Verbal Instruction

The content of verbal instruction can be redundant with the information in a static or animated diagram, or can augment or complement this information in a variety of ways. First, a verbal description provides information in a verbal code, in addition to the visual-spatial code provided by static or animated diagrams. According to dual-code theory, information represented in two codes is more memorable than information represented in a single code (Paivio, 1986). A verbal code may also enable the user to access prior knowledge about an item presented in a diagram. For example, a person may not recognize a rectangle in a highly schematic diagram as representing a cylinder

or a circle as representing a pulley. If these are labeled in the diagram or referred to by their names in the text, this might enable the user to access prior knowledge about what these components are typically made of and their typical functions (Narayanan & Hegarty, 1998).

Second, verbal instruction can direct processing of diagrams. Hegarty and Just (1993) monitored students' eye fixations while they read text accompanied by diagrams describing how pulley systems work. We examined where people switched between reading the text and viewing the diagram in the course of comprehension, and what they looked at in the diagram each time they switched. We found that people tended to switch from reading the text to viewing the diagram at the end of a clause of text, and when they switched to the diagram, they looked at the parts of the diagram that depicted the objects they had just read about in the text. The eye fixations indicated that the text was directing their processing of the diagram. Verbal direction may be particularly effective in drawing attention to areas of a diagram that are not visually salient or phases of an animation that occur quickly or involve small movements so they are unlikely to be noticed by viewing visual instruction alone (Hegarty et al., 2003; Lowe, 1999, 2003).

Verbal instruction might also be particularly important for communicating sequential information, such as the causal chain of events in a mechanical system. In complex physical systems, several objects typically move at once and there is no natural constraint on the order in which objects are fixated in a static diagram or animation of such a system. However, language has a linear structure so that language comprehension imposes a natural sequential order on information processing. This can be exploited to communicate information that has a linear structure, such as a causal chain of events.

Finally, a verbal text or commentary can provide information that is not easily communicated in static or animated diagrams. Although diagrams can provide information about the structure of a device, and its kinematics, (i.e., the geometry of movement in

a mechanical system), they are less suited to communicating about nonvisible entities such as forces (Oestermeier & Hesse, 2000). In contrast, language can express all of these types of information. Consistent with this view, when Hegarty et al. (2003) examined comprehension of static and diagrams alone and those that were accompanied by text, they found that verbal instruction was most effective in enhancing understanding of mechanisms such as a siphon, which depend on an appreciation of forces rather than kinematics alone.

In summary, verbal information can clearly augment what is learned from static and animated diagrams in a multimedia display. It may be particularly effective in drawing students' attention to information that is not perceptually salient in diagrams, communicating sequential information, and providing information about nonvisible entities, such as forces, that cannot be depicted directly in a visual-spatial display.

Format Relations Between Diagrams and Verbal Instruction

We now turn to a consideration of how the format of multimedia instruction affects learning. There are two important aspects of instructional format that must be considered and that have received considerable attention in the literature. The first concerns issues of contiguity between verbal instructions and diagrams. The second concerns the media and modalities used to deliver instruction.

CONTIGUITY

As shown in Figure 28.1, an important process in understanding multimedia is that of integrating the information presented in the text and diagrams, because the goal of comprehension is a mental model of the referent of the multimedia presentation, which is usually based on information from both media. An important step in this process is that of resolving coreference between noun phrases in the text (e.g., "the upper pulley") and diagrammatic units (e.g., a circle) that refer to the same object, in order to construct

a mental model of that referent. To accomplish this step, the verbal and diagrammatic representations must be in working memory at the same time. People sometimes ensure this by frequently interrupting their reading of a text to inspect parts of the diagram that refer to the same objects, as found by Hegarty and Just (1993). However they can also benefit significantly if verbal and diagrammatic information is presented contiguously in the multimedia display. Thus, people learn better from text and diagrams if related visual and verbal information is presented as close as possible in space rather than in separate text and diagram representations (e.g., Mayer & Gallini, 1990), and people learn more effectively from animations if they are accompanied by simultaneous commentaries rather than commentaries that are heard either before or after animations representing the same events (Mayer & Sims, 1994; Moreno & Mayer, 1999). Similarly, Faraday and Sutcliffe (1997a, 1997b) have shown that students learn more effectively from movies accompanied by commentaries if visual cues such as arrows and highlighting draw their attention to the relevant parts of the display as they are described in the commentary. Contiguity and highlighting in a display can enhance the probability that spatial and verbal information will be in working memory at the same time, so that this information can be integrated to construct a mental model of their common referent.

MEDIA AND MODALITIES

As illustrated in Figure 28.1, multimedia instruction, including diagrams and verbal instruction, can be delivered in different media and using different modalities. For example an explanation of how a machine works can be presented on a computer screen or on paper, and verbal instruction can be presented either as visual text or as an auditory commentary. To what extent do the media and modalities used to present information affect comprehension and learning? In researching this question, it is important to compare multimedia displays that

are as informationally equivalent as possible (cf. Larkin & Simon, 1987), such that they present the same information and follow the same design guidelines.

This approach was taken in a recent study by Narayanan and Hegarty (2002). They created multimedia presentations to explain two types of content, how a machine works and how specific computer algorithms work. In different presentations, they varied whether the information was presented on a computer screen and included animations, auditory commentaries and hyperlinks; or whether they were presented as printed text and diagrams. The verbal instruction in the computer and paper presentations was informationally equivalent, in that the auditory commentary presented in the computer-based instruction had the exact same words as the text in the paper-based instruction. With respect to the diagrammatic information, the computer-based instruction presented more information in that it presented a complete animation whereas the paper-based materials presented only selected frames of the animation. We examined students' comprehension of the materials and their abilities to apply this knowledge to solve novel problems. The overwhelming result was that comprehension measures were not affected by whether information was presented on a computer including animations, auditory commentaries, and hyperlinks, or on paper using traditional printed text and diagrams.

Mental Animation and Comprehension of Multimedia Displays

The fact that students have equivalent comprehension of animated and static diagrams in the context of multimedia displays again suggests that they may be able to mentally animate static displays. In the printed displays used by Narayanan and Hegarty (2002), diagrams presented the static structure of the systems while verbal instruction described the dynamic interaction between the parts. In this situation verbal instruction can augment people's natural but limited ability to mentally animate static dis-

plays by describing motions of system components that people may not be able to infer for themselves.

Mental animation may be an inherent part of the text comprehension process, as it involves converting a representation of the static structure of the device (encoded from the external diagram) to a dynamic mental model that includes a representation of how the parts of the system move to achieve its function. Consistent with this account, Hegarty and Just (1993) found that after people read a section of text describing the motion of a simple mechanical system (a pulley system), they spent several seconds inspecting a static diagram of the system that accompanied the text. Because the diagram provided no information about motion in the system, the people could not have been encoding information about motion from this diagram. The most likely interpretation of the eye-fixation behavior is that students were mentally animating the diagram to imagine the motions that they had just read about in the text. Text may be particularly effective for communicating how a system moves, because the linear structure of text can be used to guide learners through the causal chain of events that is an essential element in understanding how the system works (Narayanan & Hegarty, 1998).

Finally, asking people to mentally animate a system can enhance what they learn from multimedia. Previous research suggests that people often learn more effectively if they are more active in the learning process, and generate ideas or explanations (Chi et al., 1994; Slamecka & Graf, 1978) or first participate in activities that activate their knowledge relevant to the topic of the presentation (Britton & Graesser, 1996; Kintsch et al., 1993; McNamara, Kintsch, Songer & Kintsch, 1996). In recent studies, Hegarty et al. (2002; 2003) presented students with static diagrams of a device (a flushing cistern) and asked them to infer how the device worked (i.e. mentally animate the diagrams) before they learned about the device from multimedia presentations. Other groups of students in the same experiments learned from the multimedia presentations

without this mental animation manipulation. We found that the mental animation phase enhanced learning from multimedia such that those who engaged in mental animation were better able to describe how the system works. One interpretation of this result is that attempting to mentally animate the device allowed learners to discover what they did not understand about the device, so that they could focus on this information when viewing the multimedia displays. Consistent with this account, Rosenblit and Keil (2002) found that people overestimated their understanding of how everyday mechanical devices worked and learned more effectively when this “illusion” of understanding was shattered.

Summary and Conclusions

In summary, this chapter considers the comprehension of diagrams, animations, and multimedia presentations from an information-processing perspective. An analysis of different types of external representations revealed that these representations vary in realism or isomorphism between the representation and its referent. For example, if the referent is a dynamic process, an animation is more realistic than a series of static diagrams and if the referent is the static structure of a system, an iconic diagram is more realistic than a verbal description. More and less realistic media depend on different processes for their comprehension. For example, if the external representation of a system is more isomorphic to reality, understanding the system depends more on perceptual processes, whereas if it is more abstracted from reality, understanding the system may depend more on cognitive processes such as inference. The picture that emerges from this review is that multimedia comprehension is often an active process of knowledge construction, rather than merely a passive process of internalizing the information presented in an external display.

Because multimedia understanding depends on active information processing, in-

cluding inference and integration of different representations, it can also be influenced considerably by the abilities, skills, and knowledge of the student. We have seen that students with little domain knowledge or low spatial abilities often have difficulties with comprehension of multimedia displays. Thus what is understood from a multimedia display is jointly determined by aspects of the display and qualities of the student of a multimedia display.

The basic understanding of multimedia comprehension provided by this review can inform the design of multimedia by providing information about the effectiveness of different media for communicating different types of content. In general we have seen that iconic static diagrams can be effective for communicating the static structure of a system and also the basis for mental animation. However, comprehension of both static structure and more dynamic events can be enhanced when static diagrams are accompanied by verbal instruction, which can serve to direct readers' attention to the most relevant aspects of a diagram, guide the process of mental animation, and provide information about nonvisible entities, such as force, that cannot be represented directly in diagrammatic representations. Contrary to intuition, the literature reviewed provides no evidence that more realistic representations (e.g., animated diagrams showing movement) are more effective than more abstract representations (e.g., a series of static displays showing motion) of static displays. In contrast, traditional print media, that is, static diagrams accompanied by text, can provide highly effective external representations to aid the development of mental models of dynamic systems.

Glossary

External representation: A representation of some object, printed on paper or displayed on a computer monitor, that can be perceived by an individual.

Internal representation: A representation of some object in the mind of an individual.

Mental animation: The process of inferring how an object or physical system moves from a static display.

Mental model: The internal representation of a physical system. In other contexts, the term *mental model* has been used to denote a representation that is isomorphic to the physical situation that it represents. Although mental models of physical systems may be isomorphic to their referents, this chapter does not make a strong claim that this is always true.

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